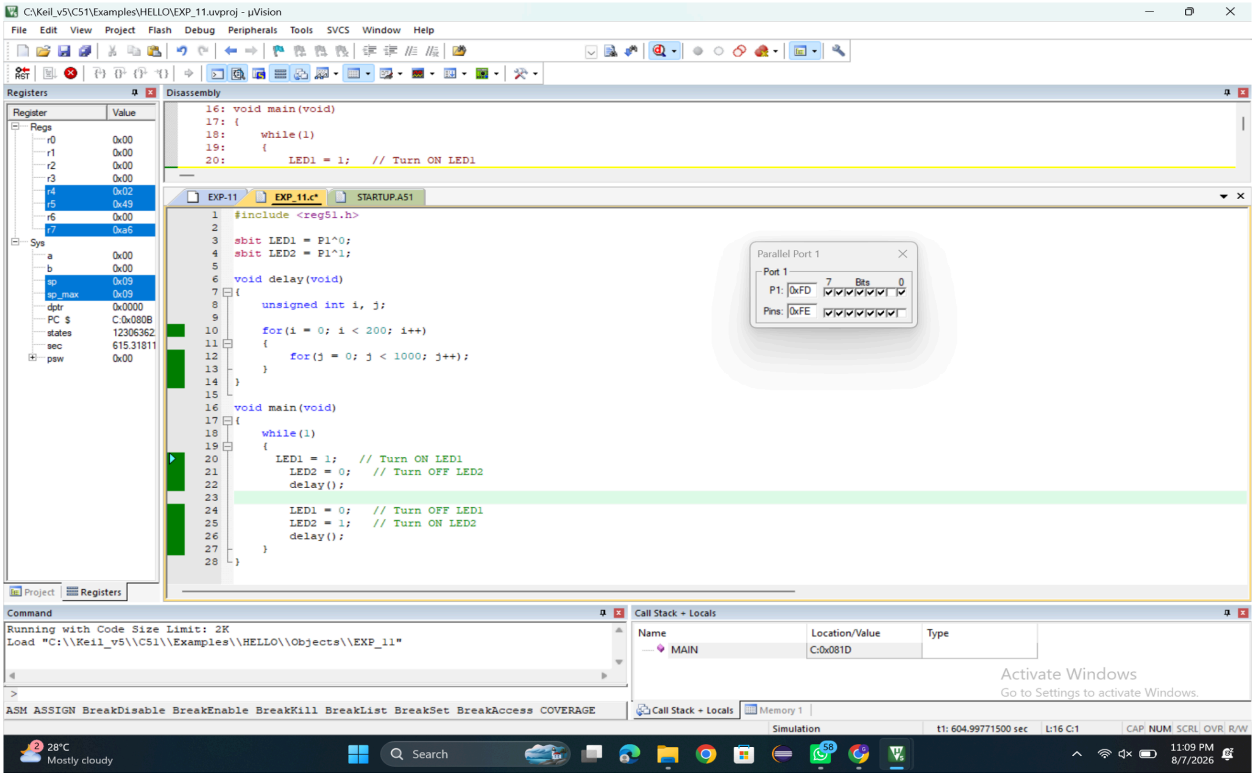
Exp No: 11 STUDY OF GPIO & BIT ADDRESSING IN AT89C51

AIM:
To study GPIO control and bit-addressable SFR using Embedded C.

SOFTWARE REQUIRED:
• Keil

PROGRAM
```
#include <reg51.h>
sbit LED1 = P1^0;
sbit LED2 = P1^1;
void delay(void)
{
   unsigned int i, j;
   for(i = 0; i < 200; i++)
   {
      for(j = 0; j < 1000; j++);
   }
}
void main(void)
{
   while(1)
   {
      LED1 = 1;   // Turn ON LED1
      LED2 = 0;   // Turn OFF LED2
      delay();
      LED1 = 0;   // Turn OFF LED1
      LED2 = 1;   // Turn ON LED2
      delay();
   }
}
```

OUTPUT



RESULT

Thus GPIO and bit addressing was studied and in Port1 the pins P1.0 and P1.1 toggled alternately.